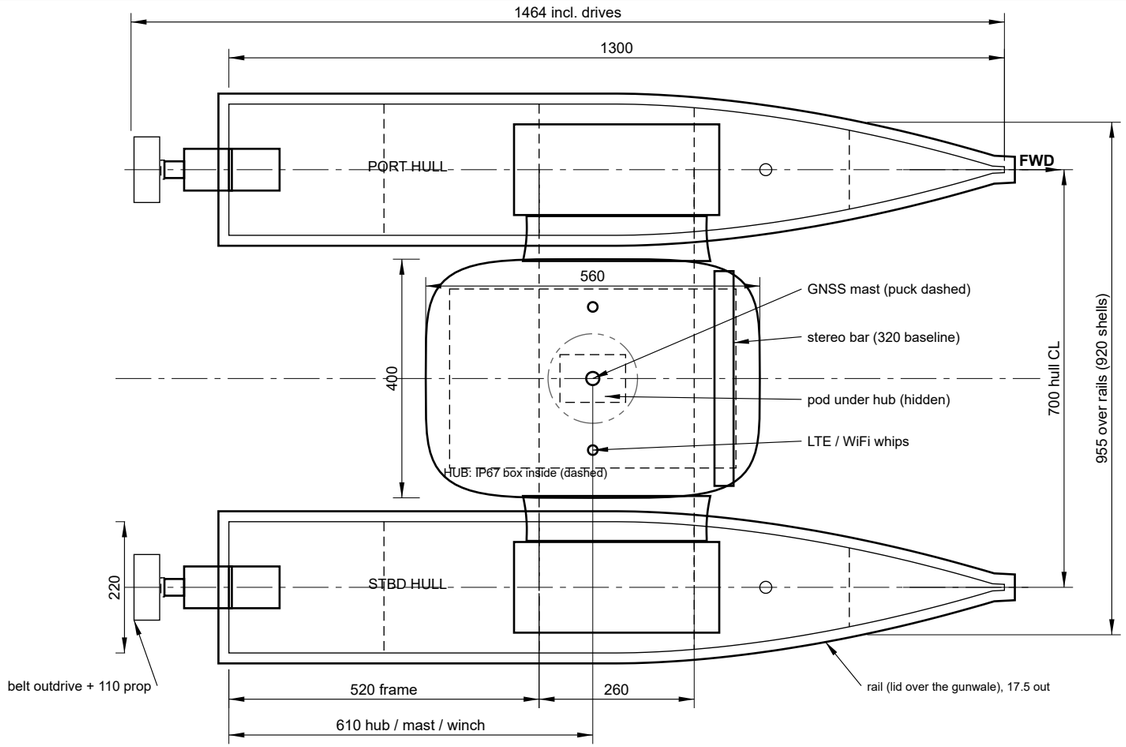
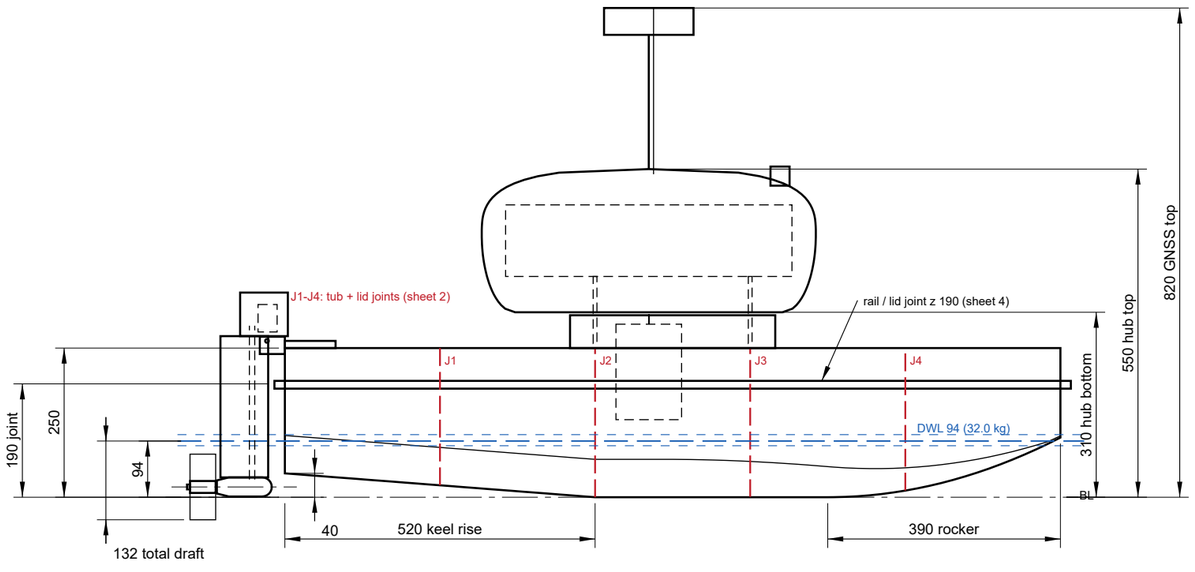


PLAN VIEW

SCALE 1:10



PROFILE (FROM STARBOARD)

SCALE 1:10

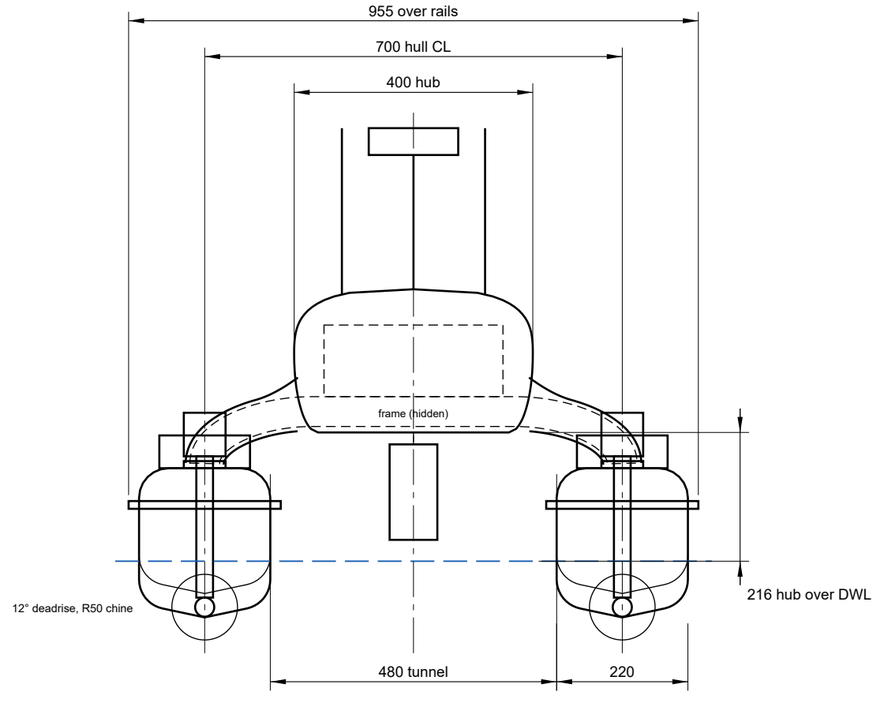
MODULES

Module	Design kg	Contents
Port tub	3.2	5 printed segments + bulkheads, flange with 37 M4 inserts, leak sensor. Bare float: no hardware
Stbd tub	3.2	mirror of the port tub
Upper module	25.6	bridge (arch frames, arms, hub: 4x M18, VESCs, Pi 5, radios, winch, pod) + 2 lids + 2 belt outdrives
Boat	32.0	incl. 10% growth margin

ASSEMBLY

1. Bolt each foot to its lid from inside: 8x M6 x 60 up through the rib bosses, EPDM pad between.
2. Bolt each outdrive to its lid's aft deck: 4x M5 from inside.
3. Press the 3 mm cord into each tub's flange groove.
4. Lower the bridge + lids onto the tubs; the drip lips locate on the tub flanges.
5. Screw down the rails: 37 M4 per hull, 60 mm pitch within 150 mm of a frame, 100 elsewhere.
6. Plug each motor cable into its arm's IP68 connector. Nothing passes through a lid or tub.

Waterlines: DWL 94 at 32.0 kg; thin blue = light 86 / max 105. Lid joint 96 above the DWL.
Lids, drives and bridge lift off as one unit; the hub keeps the electronics clear of spray.
Geometry comes from mechanical/hull_sizing.py; regenerate after changes.



VIEW FROM ASTERN

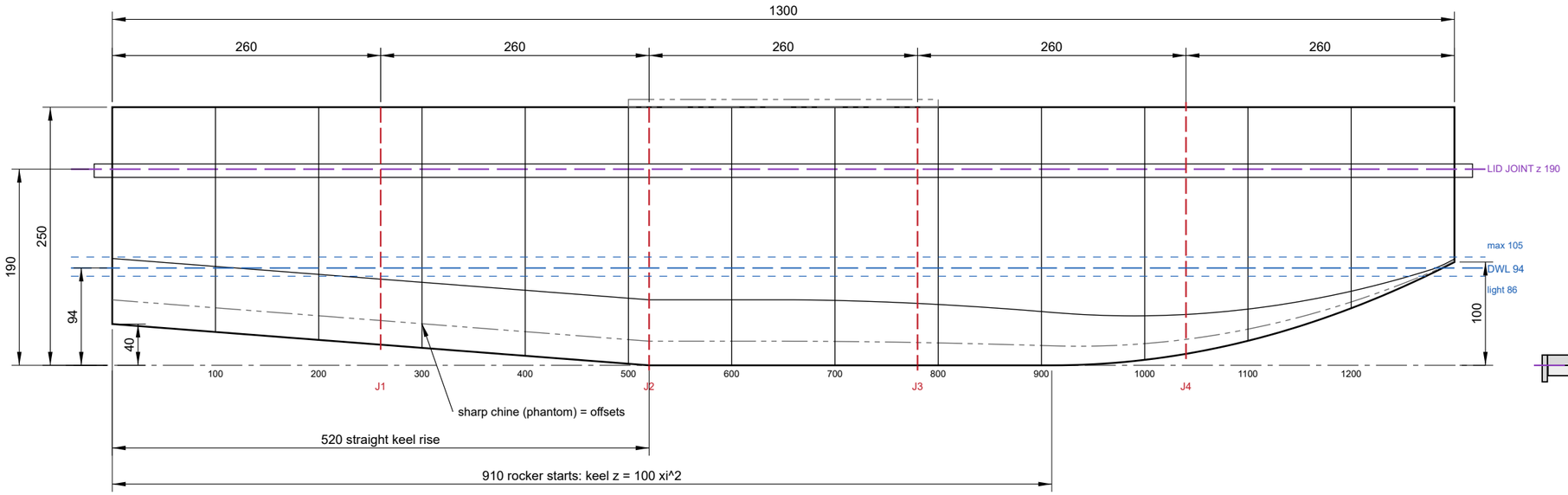
SCALE 1:10

D.R.E.A.M.S V2 — autonomous survey catamaran

General arrangement: plan, profile, view from astern

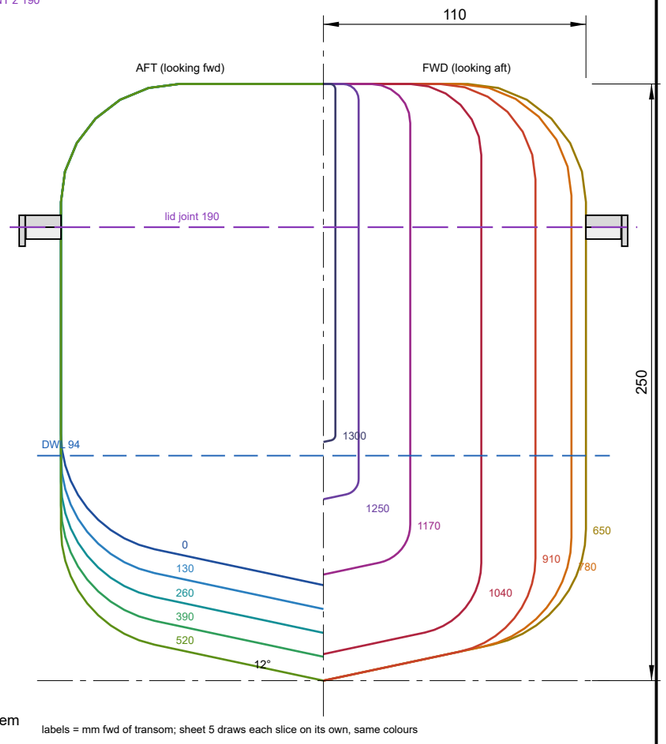
All dimensions mm unless noted. Third-angle projection. Do not scale the print; build from the numbers.

SCALE 1:10	SHEET 1 OF 5	REV C (hull lids + belt outdrive)
DATE 2026-09-27	DRAWN Claude Code for M. Hong	SRC mechanical/drawings.py



PROFILE (ONE HULL)

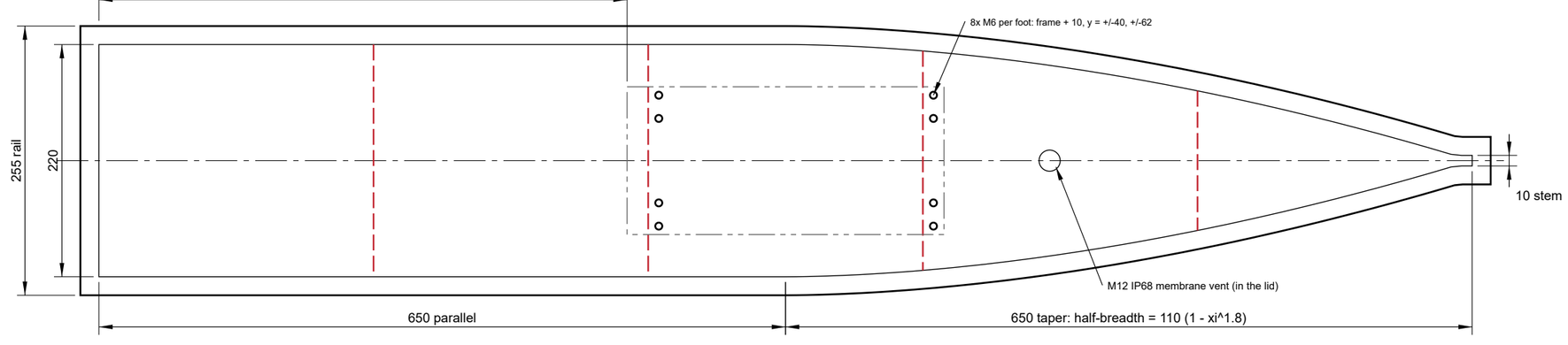
SCALE 1:5



labels = mm fwd of transom; sheet 5 draws each slice on its own, same colours

BODY PLAN (with fillets)

SCALE 1:2.5



PLAN (ONE HULL)

SCALE 1:5

OFFSETS (sharp chine; mm from transom / baseline / hull centreline)

x	keel z	chine z	half-breadth	R chine / deck
0	40.0	63.4	110.0	50 / 50
130	30.0	53.4	110.0	50 / 50
260	20.0	43.4	110.0	50 / 50
390	10.0	33.4	110.0	50 / 50
520	0.0	23.4	110.0	50 / 50
650	0.0	23.4	110.0	50 / 50
780	0.0	22.1	103.9	47 / 47

x	keel z	chine z	half-breadth	R chine / deck
910	0.0	18.9	88.9	40 / 40
1000	5.3	21.0	73.9	33 / 33
1100	23.7	35.1	53.3	24 / 24
1170	44.4	52.2	36.4	16 / 16
1250	76.0	79.1	14.8	7 / 7
1300	100.0	101.1	5.0	2 / 2

CONSTRUCTION (open tub + removable lid)

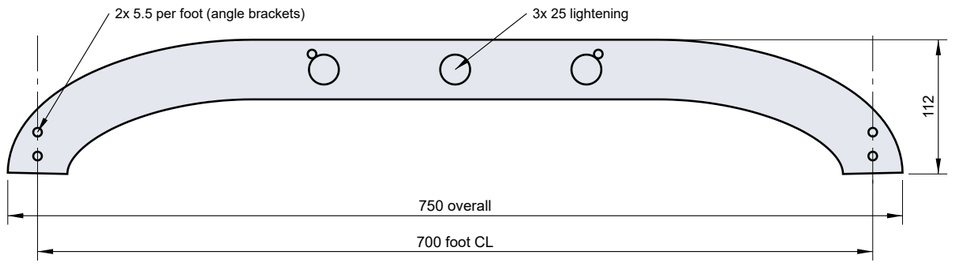
- Loft the sharp-chine hull from the offsets, fillet chine R50 and deck edge R50 (0.45 x half-breadth near the stem), then split the body at the lid joint plane z 190: tub below, lid above.
- PETG or ASA, 2.0 wall. Glass (200 g/m2 + epoxy) the tub and the lid separately, up to the flanges.
- Tub: 5 x 260 segments, each printed on its aft 2.4 bulkhead; bulkheads stop 5 under the joint plane.
- Lid: 5 x 260 segments printed on end; lid segments 3 and 4 stand on their 8 mm ribs (J2 / J3) with the foot-bolt bosses.
- Joints J1-J4: 8 mm half-lap spigots on the tub and on the lid (sheet 3).
- Rail: 15 mm flanges, 3 mm cord in a 5 x 2.2 groove, 37x M4 into M4 x 8.1 heat-set inserts in 10 mm bosses under the tub flange (sheet 4).
- One sealed space per hull: leak sensor at its lowest point (Boat PCB J15 / J16), M12 vent in the lid.
- Shell 4.1 kg per hull incl. bulkheads, ribs and stiffeners, + 1.0 kg rail.

D.R.E.A.M.S V2 — autonomous survey catamaran

Demi-hull lines, lid joint plane, print segments, offsets and fillets

All dimensions mm unless noted. Third-angle projection. Do not scale the print; build from the numbers.

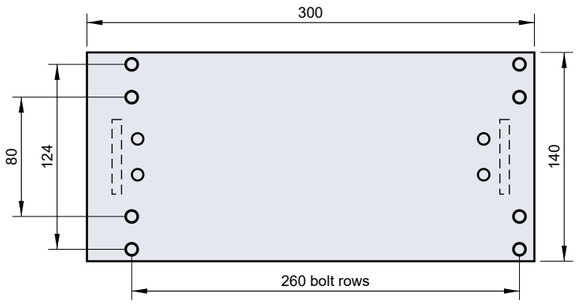
SCALE 1:5 / 1:2.5	SHEET 2 OF 5	REV C (hull lids + belt outdrive)
DATE 2026-09-27	DRAWN Claude Code for M. Hong	SRC mechanical/drawings.py



1/4 in (6.35) 6061-T6, 2 req. Section 50 deep, +/-25 from the arch centreline; feet stand on the foot plate, 7.5 above the lid deck. Cut file: cad/arch_frame_plate.dxf

ARCH FRAME PLATE (FROM AHEAD)

SCALE 1:5



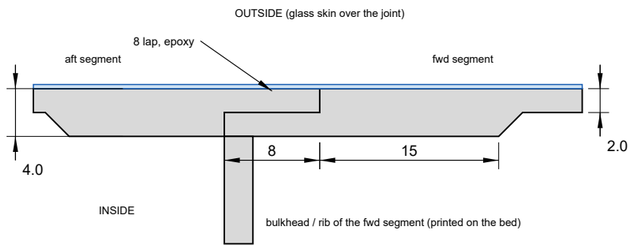
Arch centreline (stbd arm)	y	z
hub span, straight	0 to 170	345
Bezier P0 (arm root)	170	345
Bezier P1	260	345
Bezier P2	350	308
Bezier P3 (foot)	350	257.5

Port arm is the mirror image.
Tangent horizontal at P0, vertical at P3.

6 mm 6061. Frames (hidden) at -130 / +130. 8x M6 tapped through at frame + 10 (bolts come up from inside the lid); 2x 5.5 per angle bracket

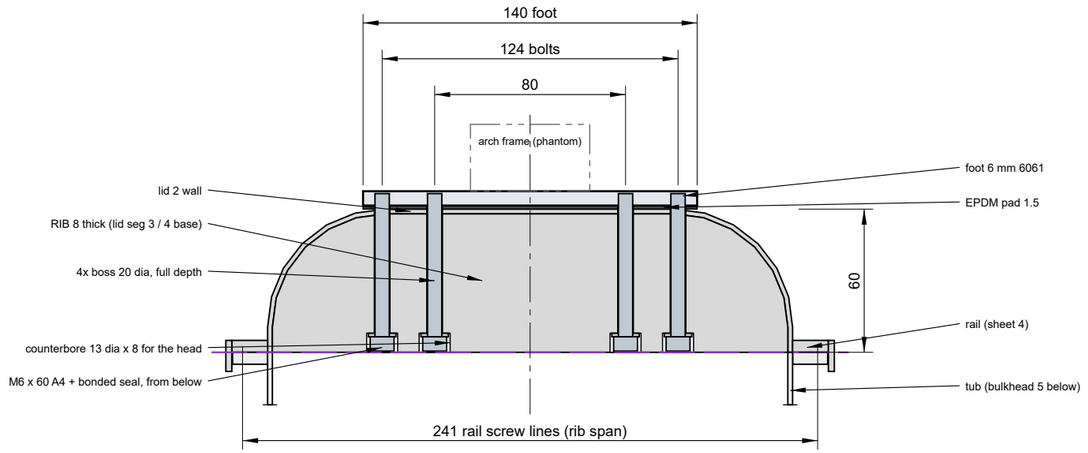
FOOT PLATE (PLAN, 2 req)

SCALE 1:4



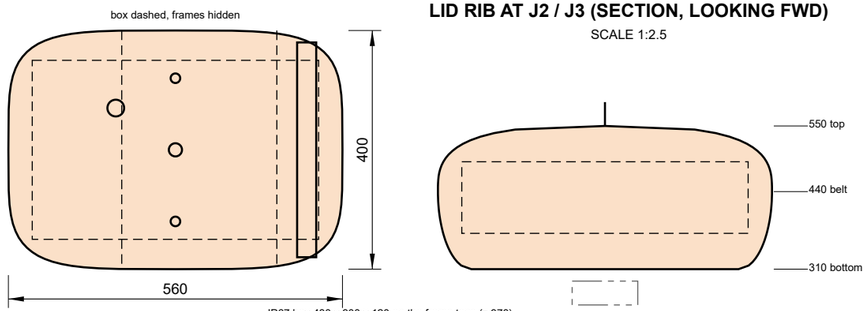
SEGMENT JOINT, TUB + LID (45° chamfers print unsupported)

SCALE 2:1



LID RIB AT J2 / J3 (SECTION, LOOKING FWD)

SCALE 1:2.5



IP67 box 480 x 300 x 120 on the frame tops (z 370)

HUB COWL LAYOUT (PLAN + SIDE)

SCALE 1:10

HARDWARE (whole boat)

Qty	Item
2	Arch frame plate, 1/4 in 6061-T6, laser / waterjet cut (DXF)
2	Foot plate, 6 mm 6061, 300 x 140, 8x M6 tapped
4	Angle bracket 25 x 25 x 3 Al, 40 long (frame to foot)
16	M6 x 60 A4 socket cap + M6 bonded seal washer (feet, from inside)
2	EPDM pad 1.5 mm, 300 x 140 (under each foot)
74	M4 x 12 A4 pan head + washer, and M4 x 8.1 brass heat-set insert (rails)
2	3 mm EPDM / silicone cord, 2.85 m (one loop per tub)
1	Hub cowl, 2 arm fairings, 2 foot shoes (printed PETG / ASA)
1	IP67 box ~480 x 300 x 120, glands, 2 IP68 motor connectors
2	Belt outdrive, complete (sheet 4)
2	M12 IP68 membrane vent (one per lid)
2	Leak sensor at each tub's lowest point (Boat PCB J15 / J16)

LOAD PATH

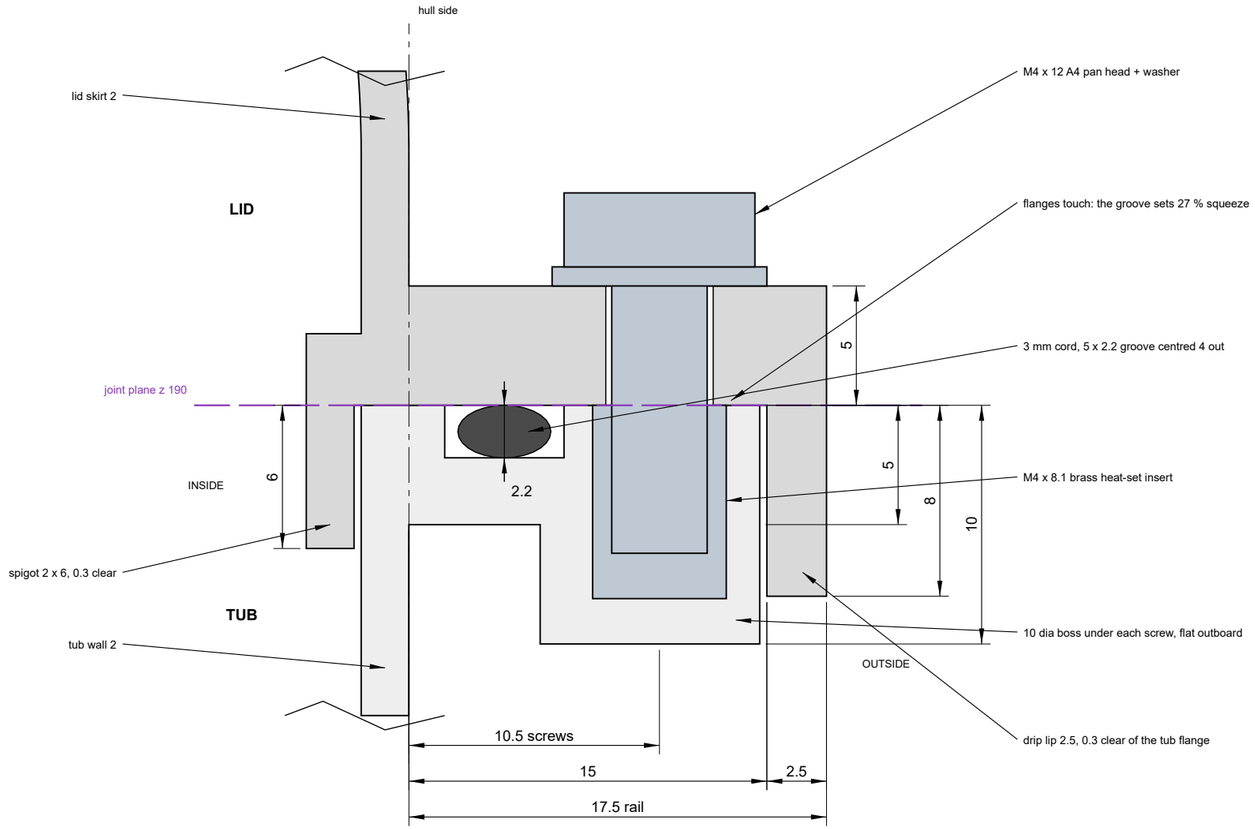
Each frame is a portal fixed at its feet.
Worst case: hulls pitching out of phase, a_cg 2 g
136 N m per frame -> 52 MPa,
SF 5.4 on 6061-T6 yield.
Foot: pivots on its far edge; the 8
M6 in the rib bosses hold it: SF 3.5.
Lid rib carries it to the rail: SF 4.7.
Rail: 3 M4 inserts per frame: SF 4.7.
Nothing goes through a tub: all
hardware bolts to the lids.
Details: docs/hull_and_crossbeam_sizing.md

D.R.E.A.M.S V2 — autonomous survey catamaran

Arch frame, foot plate, lid rib, hub layout, segment joint

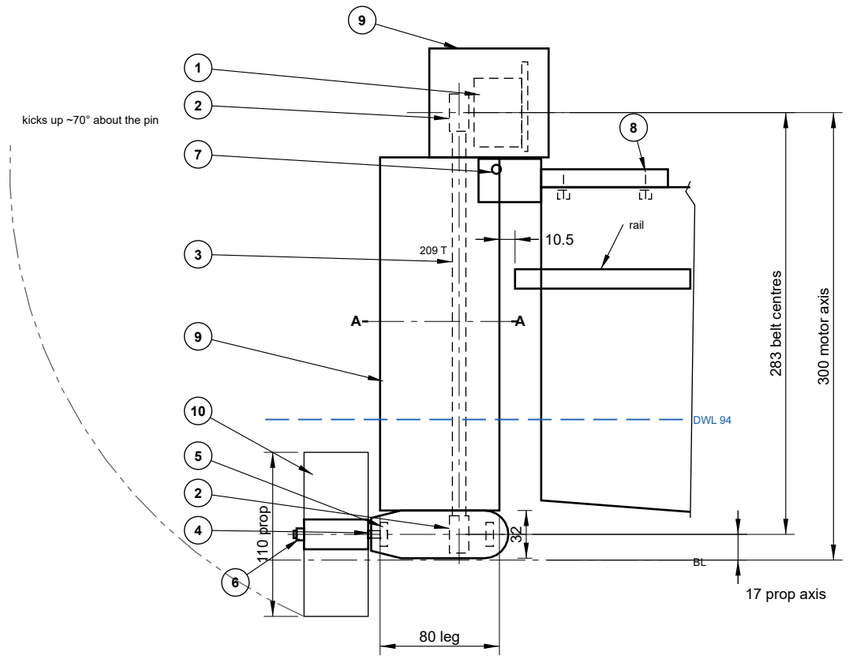
All dimensions mm unless noted. Third-angle projection. Do not scale the print; build from the numbers.

SCALE as noted	SHEET 3 OF 5	REV C (hull lids + belt outdrive)
DATE 2026-09-27	DRAWN Claude Code for M. Hong	SRC mechanical/drawings.py



OUTDRIVE NOTES

- The leg and pod free-flood, so the belt and both bearings run wet: stainless bearings, a PU belt with aramid or stainless cords, and a fresh-water rinse after use.
- Tension the belt by sliding the motor up its slotted plate. The whole drive tilts about the pin, so a kick-up doesn't change the tension.
- Forward thrust holds the leg on its rubber stop; the detent holds it against reverse thrust (differential steering) and lets go when the leg hits something.
- The leg is split on its chord plane so the belt goes in whole. Motor wires run up the arm fairing to the hub; nothing passes through the lid.
- Stbd prop RH, port prop LH (tops turn outboard). A 9 mm belt is near its rating at full speed; cruise is well inside it.

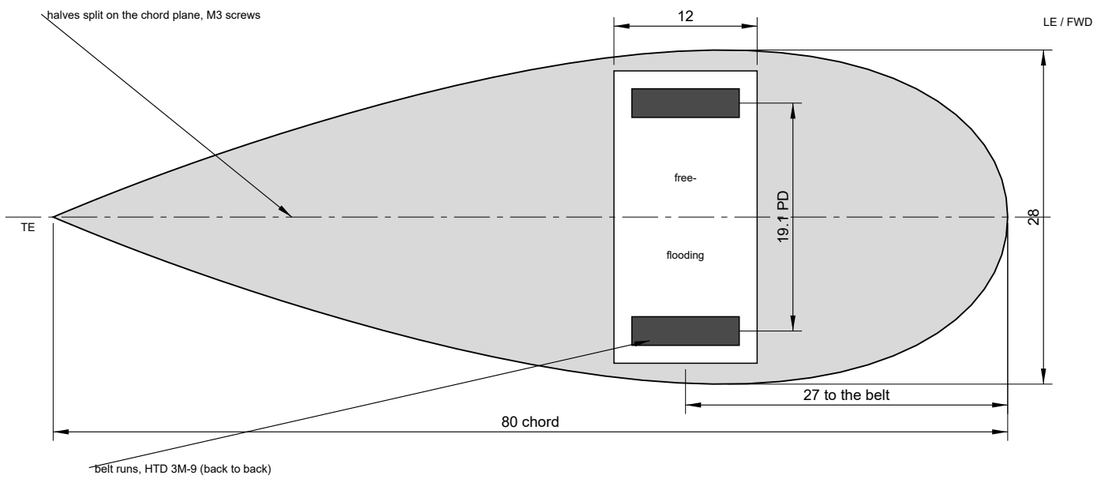


BELT OUTDRIVE (FROM STARBOARD)

SCALE 1:4

RAIL: LID OVER THE GUNWALE (SECTION THROUGH A SCREW)

SCALE 4:1



SECTION A-A (THROUGH THE LEG)

SCALE 2:1

OUTDRIVE PARTS (per drive; 2 req)

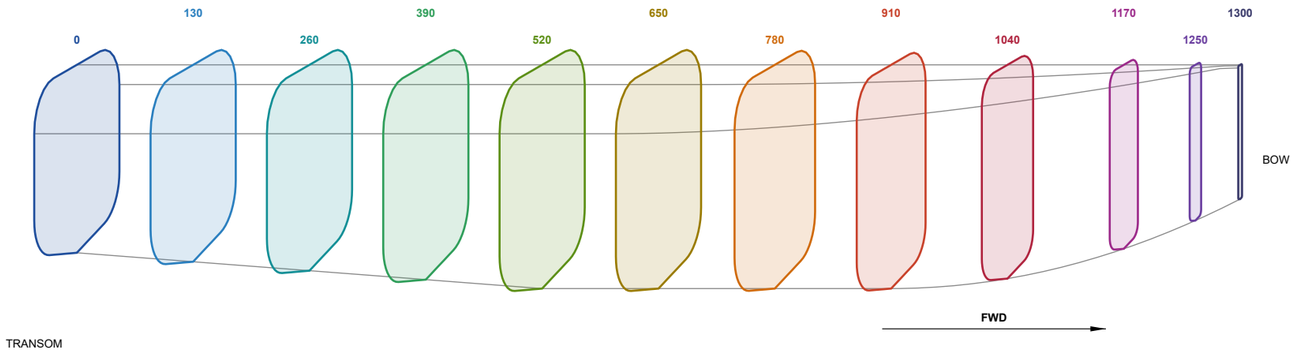
No.	Qty	Item
1	1	Outrunner ~380 KV, 4108 size, >= 25 A (e.g. SunnySky X4108S-380KV), on the slotted motor plate
2	2	HTD 3M pulley, 20T (19.1 PD), for a 9 mm belt, 5 mm bore + set screw
3	1	HTD 3M belt, 9 mm, 209 T (627 mm); 15 mm belt + pulleys for long runs at 3 m/s
4	1	316 stainless shaft 5 mm, D-flat under the prop, M5 thread for the nylock
5	2	625-2RS bearing, stainless (5 x 16 x 5)
6	1	M5 nylock A4 + washer
7	1	Tilt pin M6 A4 + 2 flanged bushings; spring-plunger detent that holds reverse thrust
8	4	M5 x 20 A4 + washers: bracket to the lid deck, from inside the lid
9	1	Printed in ASA: bracket, tilt head, cover, leg halves (M3 screws), pod
10	1	Prop 110: props/toroidal_3loop_p75_RH_shaft (stbd) / _LH_shaft (port), or a 110 mm RC prop

D.R.E.A.M.S V2 — autonomous survey catamaran

Hull lid joint (rail) and belt outdrive

All dimensions mm unless noted. Third-angle projection. Do not scale the print; build from the numbers.

SCALE as noted	SHEET 4 OF 5	REV C (hull lids + belt outdrive)
DATE 2026-09-27	DRAWN Claude Code for M. Hong	SRC mechanical/drawings.py



WHERE EACH SLICE IS (ONE HULL)

SCALE sketch, not to scale

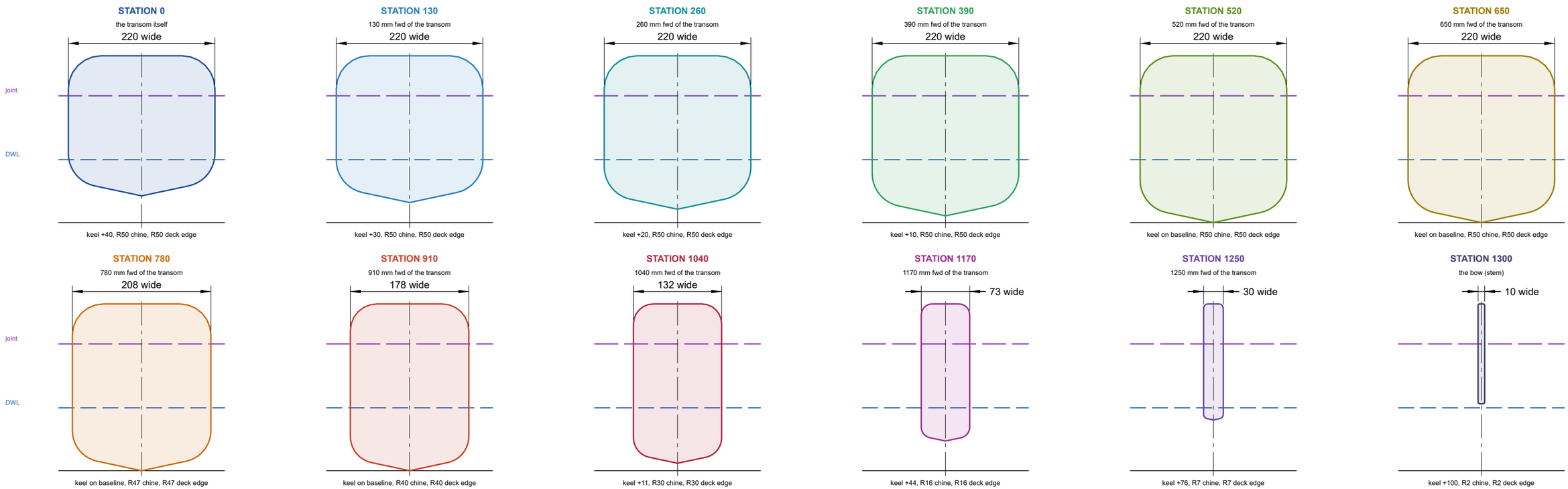
HOW TO READ THE BODY PLAN (SHEET 2)

Each coloured shape is one slice straight across ONE hull, like a slice of bread. Its number is where the slice is: mm forward of the transom.

Sheet 2 stacks all the slices on one drawing. The hull is the same on both sides, so it draws only half of each slice: the stern half of the boat on the left of the centreline, the bow half on the right, in these colours.

Reading from stern to bow:
- 0 to 520: the same shape, but the keel lifts toward the transom (+40 mm at 0), so the stern draws less water.
- 520 to 650: the widest part (220 mm), keel on the baseline.
- 650 to 1300: it narrows toward the bow; from 910 the keel also rises, to +100 at the bow, which ends in a 10 mm flat face.

In each box: blue dashes = waterline (94), purple = lid joint (190).
In SolidWorks, each box is one sketch on a plane at that x (guide, A2).



Scale 1:6. Full slices, both halves, looking forward from the stern.

D.R.E.A.M.S V2 — autonomous survey catamaran

Hull sections one at a time: a reading aid for the body plan

All dimensions mm unless noted. Third-angle projection. Do not scale the print; build from the numbers.

SCALE 1:6 / sketch	SHEET 5 OF 5	REV C (hull lids + belt outdrive)
DATE 2026-09-27	DRAWN Claude Code for M. Hong	SRC mechanical/drawings.py